EXAMFORGE — AI-POWERED TEST GENERATION PLATFORM

Technical Architecture, Engineering Specification & Platform Evaluation Report

Project Name: ExamForge Enterprise	**Release Version:** 2.4.0 Production	**Target Audience:** Educators & Academies	**Audit Date:** October 2026
Architecture: Hybrid MVC + EDA	**Frontend Stack:** Next.js 16 (React 19)	**Data Infrastructure:** Supabase (PG) + SQLite	**AI Engine:** Google Gemini Flash/Pro

1. Executive Summary & Problem Formulation

Preparing high-stakes secondary and higher-secondary examinations (Classes 9 and 10) traditionally presents severe operational bottlenecks. Educators face excessive manual drafting overhead, curriculum drift, cognitive imbalance across Bloom's Taxonomy, and heightened vulnerability to examination leaks. Traditional authoring methods lack programmatic validation for mark-to-time distributions, leading to inconsistent evaluations.

ExamForge addresses this through a high-performance web platform combining a validated 2,100+ item PostgreSQL curriculum question repository (Supabase) with real-time generative AI synthesis (Google Gemini). Instructors define deterministic blueprints, calibrate difficulty tiers (Easy/Medium/Hard), generate multi-variant anti-cheat sets, and produce publication-ready Word (.docx) and high-fidelity PDF exam papers within seconds.

2. Hybrid Architecture (MVC + Event-Driven Architecture)

ExamForge implements an enterprise Hybrid MVC + EDA pattern, guaranteeing sub-second response times, modular maintainability, and non-blocking asynchronous event handling:

Layer	Technical Responsibilities	Core Components & Technologies
Model Layer	Encapsulates schema validation, lookup caching, and dual-database queries. Isolates remote bank from local persistence.	Prisma ORM, SQLite (local papers/sessions), Supabase Client (curriculum hierarchy & questions).
Controller Layer	Orchestrates test generation, blueprint validation, batch partitioning, deduplication, and auth.	TestController, QuestionController, AnalyticsController, BlueprintSuggester, AuthController.
Event Layer (EDA)	Process-wide asynchronous EventEmitter bus managing telemetry, audit logging, and cache invalidation.	EventBus (emits TEST_GENERATED, CACHE_INVALIDATED, TELEMETRY_RECORDED, AUTH_SUCCESS).
View Layer	Distraction-free, responsive UI with stateful wizard, question bank explorer, preview studio, and AI chat.	Next.js 16 (App Router, Turbopack), React 19, Tailwind CSS v4, Radix UI Primitives, KaTeX.

Core Architectural Advantages: Zero-latency decoupled execution via EventBus; genuine OpenXML .docx compiler producing 2-column examination papers; and Gemini Flash AI synthesis with mathematical KaTeX formatting.

3. Four-Stage Guided Exam Engineering Workflow

ExamForge features a streamlined, 4-step wizard that guarantees 100% curriculum adherence while eliminating cognitive overload for teachers:

Stage	Educator Interaction	System Action & Validation
Step 1: Class Selection	Chooses grade level (Class 9 or 10) via live stats cards showing subject counts and total bank volume.	Validates class ID, queries Supabase curriculum hierarchy, and resets downstream caches.
Step 2: Subject Discovery	Selects course (Computer Science, Physics, Mathematics, Biology, Chemistry).	Loads all associated chapters, subtopics, and total question inventory instantly.
Step 3: Granular Syllabus	Selects or deselects specific chapters and fine-grained subtopics with Select All shortcuts.	Calculates exact topic IDs for targeted query filtering, enforcing minimum 1 chapter selected.
Step 4: Blueprint Setup	Configures question counts, marks per question, and difficulty breakdown (Easy / Med / Hard).	Runs blueprint matrix generator, validates question availability, and synthesizes test paper.

4. Cognitive Calibration & Sectional Distribution Matrix

Exam papers are structured according to official academic assessment standards across 3 discrete sections with strict cognitive controls:

Section	Question Type	Cognitive Target (Bloom's)	Assessment Mechanism
SECTION A	Multiple Choice (MCQs)	Remembering & Understanding (Recall, Terms)	1 Mark/Q. Auto-graded with randomized distractor shuffling.
SECTION B	Short Answer Questions	Applying & Analyzing (Concepts, Calculations)	2–4 Marks/Q. Evaluated against model answers & rubrics.
SECTION C	Long Descriptive Essay	Evaluating & Creating (Synthesis, Proofs)	5–10 Marks/Q. Comprehensive analytical rubric grading.

5. Multi-Variant Batching & Anti-Cheat Engine

To eliminate examination cheating and paper leaks in shared exam halls, ExamForge features an algorithmic Multi-Variant Batch Generator. Instructors can synthesize up to 6 parallel disjoint versions (Set A, B, C, D, E, F) simultaneously. The engine enforces mutual exclusion so that each variant contains completely unique questions, or applies cryptographic pseudo-random question and option shuffling while holding total marks, time allowed, and difficulty ratios identical.

6. Interactive Studio & Multi-Format Export Engine

Following generation, educators enter the Interactive Test Studio. Features include: (1) Drag-and-drop question reordering; (2) Inline question editing; (3) Single-click AI question regeneration; (4) Instant AI answer explanations; (5) Practice Online Mode with countdown timer and instant scoring; (6) Institution branding with academy name, logo upload, watermark, roll number box, and detached student Answer Key attachments; and (7) Genuine OpenXML .docx and vector PDF export.

7. Database Schema & Data Relational Architecture

ExamForge operates a dual-database persistence tier ensuring sub-second throughput, offline development capability, and scalable remote storage:

Table / Entity	Storage Engine	Schema Definition & Relationships
classes & subjects	Supabase (PG)	Root curriculum taxonomy; stores class grades (9th, 10th), subject codes, and course credit metadata.
chapters & topics	Supabase (PG)	Hierarchical course syllabus nodes with numeric sequence sorting and question bank indexing.
questions & options	Supabase (PG)	2,100+ vetted curriculum items with stem text, options A-D, Bloom's level, source tags, and model answers.
tests & user_sessions	Prisma (SQLite)	Full JSON blueprint snapshots, test variants, user accounts, and authentication cookies.

8. Distraction-Free Focus Mode & Universal Access Architecture

ExamForge features dedicated full-window focus modes for Test Generation and AI Chat. When entering these workflows, global headers, footers, and floating widgets are cleanly suppressed to maximize cognitive focus. Navigation is anchored by a persistent sticky focus bar providing clear 'Back to Dashboard', 'Back to Previous Step', and Theme Toggle controls. Furthermore, artificial role gating has been eliminated: all registered educators and students possess universal authoring, batch generation, saving, and deletion permissions.

9. Quality Assurance, Build Stability & Performance Metrics

The codebase has undergone exhaustive QA audits and regression testing to ensure enterprise-grade stability:

Audit Dimension	QA Specification & Target	Achieved Benchmark & Verification Status
Turbopack Production Build	Clean zero-error Next.js 16 build	100% Pass (0 errors, 0 lint warnings, optimized static routes).
Query & Generation Latency	Test synthesis under 1,500ms	Avg 480ms for bank queries; ~2.1s for hybrid Gemini AI fills.
Export File Integrity	100% compliance with OpenXML DOCX	Passes Word 2016-2024 & Office 365 validation without repair prompts.
Error Boundary Containment	Isolated widget failure recovery	Zero global crashes; graceful fallback toasts and local retry buttons.

10. Strategic Roadmap & Future Horizons

The project roadmap includes: (1) Automated Optical Mark Recognition (OMR) camera grading for mobile assessment; (2) Multi-lingual paper synthesis supporting Urdu, Arabic, and regional curricula; (3) Bidirectional Learning Management System (LMS) synchronization with Google Classroom, Canvas, and Moodle; and (4) Predictive student performance analytics identifying institutional learning gaps.

DOCUMENT STATUS	LEAD ARCHITECT	COMPLIANCE AUDIT
Approved for Production Deployment	**ExamForge Core Engineering Team**	**ISO / Academic Board Examination Standards**